Problem statement:

Develop a machine learning model to forecast hazardous flux levels of >2 MeV energetic electron at geostationary orbit using solar wind parameters. The model should provide an early warning 1 hour in advance whenever the predicted electron flux is expected to exceed the 2 MeV threshold. The solution will use 5 years (2015–2019) of GOES electron flux data and corresponding Wind spacecraft solar wind data, with the goal of achieving at least 90% prediction accuracy.

Data Collection:

Decided to use 1 min averaged datasets with timeline **2011/01/01 00:00:00.000 to 2020/01/01 00:00:00.000 (9 years)** as during this timeframe, there were multiple solar events which is necessary to diversify the dataset and to make accurate predictions.

1. For Wind Data **OMNI_HRO_1MIN Dataset**:
   Parameters Selected:
      - Magnitude of avg. field vector (nT)
      - Bx (nT), GSE
      - By (nT), GSM
      - Bz (nT), GSM
      - Flow Speed (km/s), GSE
      - Proton density (n/cc)
2. For GOES (Electron Flux Data)
   GOES15_EPEAD-SCIENCE-ELECTRONS-E13EW_1MIN
   Parameters Selected:
   - A/W Detector: Electron integral flux E2 >2 MeV (background, contamination and dead time corrected)

Data Preprocessing

- Decided to use the “Parquet” format as the primary way to store the data as the dataset is huge around 2.6 million records. Therefore the two data files (goes.cdf and omni.cdf) are converted into goes.parquet and omni.parquet.
- As NASA uses dummy data like `-1.0E+31` or `9999.0` in the dataset where the values are missing. We have used *“fillval_to_nan=True”* to convert these missing values into standard NaN (Not a Number) values, used in numpy and pandas.
- Combined data of both files (omni.parquet and goes.parquet) on the timestamp field which NASA calls/writes as *“Epoch”*.
- The merging of data revealed a finding; the *Omni* dataset was having 2628001 rows and the *Goes* dataset was having 2625121 rows. So this is a complete 2 days missing data.
- As we have merged the dataset on the timestamp so the final merged dataset is also having 2 days of data missing. Leading to the dataset accuracy to 99.89%

- Due to this issue now the dataset has alignment issues, i.e. the gap between two records can be more than 1 minutes depending on how many records are missing in between these two records.
- To solve this problem I used the re-indexing of the time(i.e. epoch) column. Which means having a continuous time interval of 1 min and for the missing values and records the values will be NaN.
- After re-indexing i accurately have 2628001 rows, out of which the missing rows have just the timestamp and other parameters as NaN.
- Now the next issues is the no. of missing missing values the merged dataset has

Column	No. of NaNs	Percentage of NaNs
F	191945	7.303840
BX_GSE	191945	7.303840
BY_GSM	191945	7.303840
BZ_GSM	191945	7.303840
flow_speed	614867	23.396757
proton_density	614867	23.396757
E2W_COR_FLUX	656310	24.973735

- Based on this analysis, 7 to 8 percent of missing values are acceptable, but the parameters [flow_speed, proton_density, E2W_COR_FLUX] has very large missing data ~25%
- So, to reduce this missing data I need to Interpolate data.
- I have decided to use a mix of two Interpolation techniques i.e. liner interpolation and forward filling interpolation.
- As solar wind movement and parameters do not change very drastically within very less time. so,I have decided to use linear interpolation for gaps <= 15 mins and forward filling interpolation for gaps >15 & <=30 mins. And for the gaps > 30 mins I will leave it as NaNs only, otherwise it will degrade the quality of the dataset.
- Also I am not going to perform Forward Filling Interpolation on E2W_COR_FLUX as it is my output variable.
- So, after interpolation missing values analysis shows:

Column	No. of NaNs	Percentage of NaNs
F	68825	2.618911
BX_GSE	68825	2.618911
BY_GSM	68825	2.618911
BZ_GSM	68825	2.618911
flow_speed	71762	2.730669
proton_density	71762	2.730669
E2W_COR_FLUX	513650	19.545274

Feature Engineering:

- **F:** Magnitude of Interplanetary Magnetic Field (IMF), unit = nanoTesla (nT)
- **BX_GSE:** Magnetic field in the X direction.
- **BY_GSM:** Magnetic field in the Y direction.
- **BZ_GSM:** Magnetic field in the Z direction.
- **Flow_Speed:** Solar wind (Plasma) speed, unit = km/s
- **Proton_Density:** unit = particles/cm^3 (n/cm^3)
- **E2W_COR_FLUX:** measure the elector flux of electron energy > 2MeV (n/cc)
- **Log_Flux: log10(flux+1)** used to compress large numbers
- **Target_Log_Flux_1H**: Target variable i.e Log_Flux after 1 hour from current time
- **Lag Features (15m, 30m, 60m):** Log_Flux, F, BX_GSE, BY_GSM, BZ_GSM, Flow_speed, Proton_Density
- **Rolling Features: standard deviation (1h) and mean(1h and 3h):** BZ_GSM, Flow_Speed, Log_Flux

Model Selection:

- **Model Selection Technique:** Walk-Forward CV (Cross Validation)
- Algorithms:
    a. Random Forest
    b. XGBoost
    c. LightGBM
    d. CatBoost
- findings: